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Science with the Liverpool Telescope
Assignment Two
Classification of Stars

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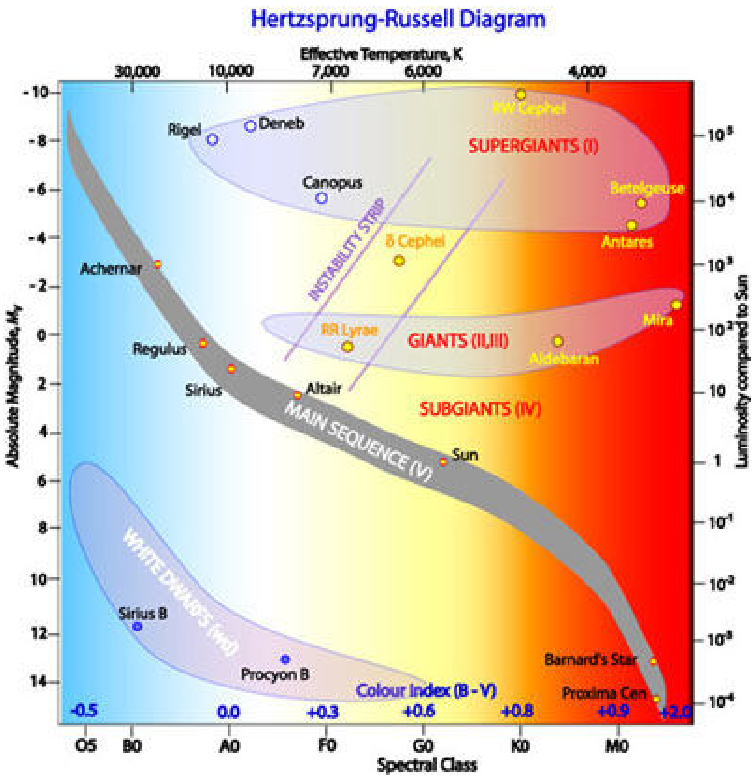

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*Image Credit : R. Hollow, CSIRO

1 Introduction

This assignment is to classify the stars, based on spectra taken by the SPRAT of the Liverpool Telescope. The data sets include 9 observations, i.e. Target Star 1 to 9, and 2 Standard Calibration Stars a and b, with spectra taken immediately after the target spectra, using the same instrument, i.e. SPRAT. The Standard Calibration Stars used are located as close to the Target Stars as possible.

As each observation has been taken in different atmosphere environments, calibration is needed to remove the effects of the changing atmosphere.

With the calibrated spectra, the next stage is to classify the spectral type of the stars. A systematic approach is used to compare the calibrated spectrum of each Target Star to a range of spectra of stars, drawn from "the library of stellar spectra" by Jacoby, Hunter and Christian (1984), where classification of the those star is known. Finally, the type of the Target Stars are determined with reasoning.

2 Calibration

As each wavelength of light is affected by the atmosphere in a different way, the spectra of the Standard Calibration Stars, taken immediately after the Target Stars, are used to compare with their "ideal" calibrated spectra and work out the ratios. And those ratios are multiplied with the spectra of the Target Stars, arriving at the calibrated spectra of the Target Stars. As those Standard Stars have been very carefully studied, their "ideal" fully-calibrated spectra are known. Another calibration method, i.e. arithmetical difference between the Target Stars and Calibration Stars has been examined but found that the results were too far off.

Here is a fully worked example for Star 1 at wavelength 4100 Angstrom :

- Star 1 at wavelength 4100 Angstrom has a normalized flux of 0.0816750, which was taken on 2015-09-06
- The Standard Calibration Star a at the same wavelength has a normalized flux of 0.2547557
- The Standard Calibration Star a has a known "ideal" calibrated flux of 2.6529820
- The ratio between the ideal calibrated flux and normalized flux of the Calibrated Star a = 2.6529820 / 0.2547557 = 10.4138278
- The calibrated flux of Star 1 = 10.4138278 $\times$0.0816750 = 0.8505493

See the following figures for all un-calibrated and calibrated spectra for Star 1 to 9. The spreadsheet for each detailed calculations is also attached to this assignment.

- Star 1 - calibrated with Standard Calibration Star a

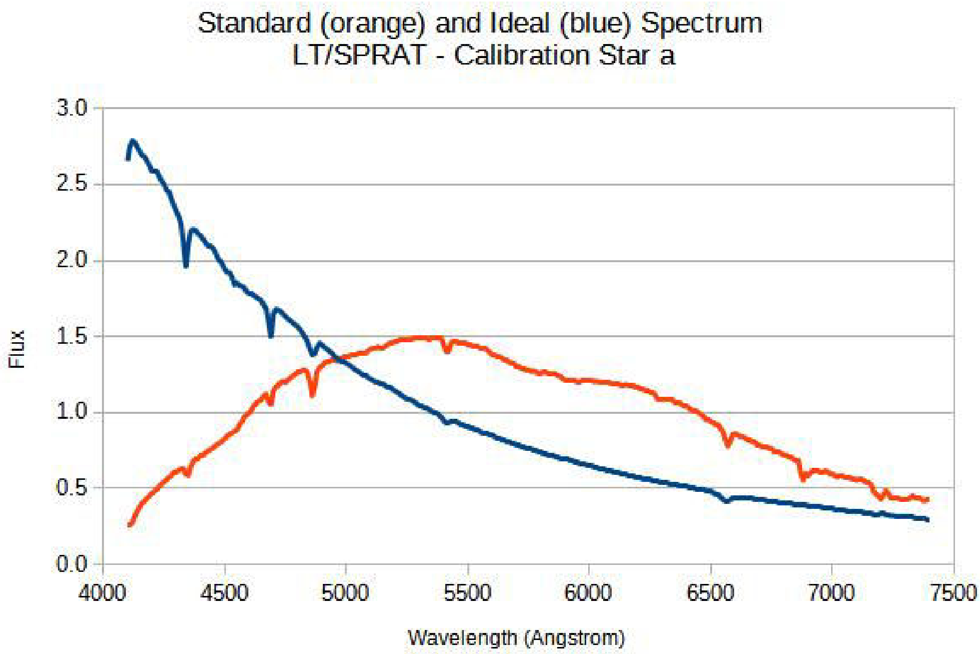

Figure 1: Calibration Star a

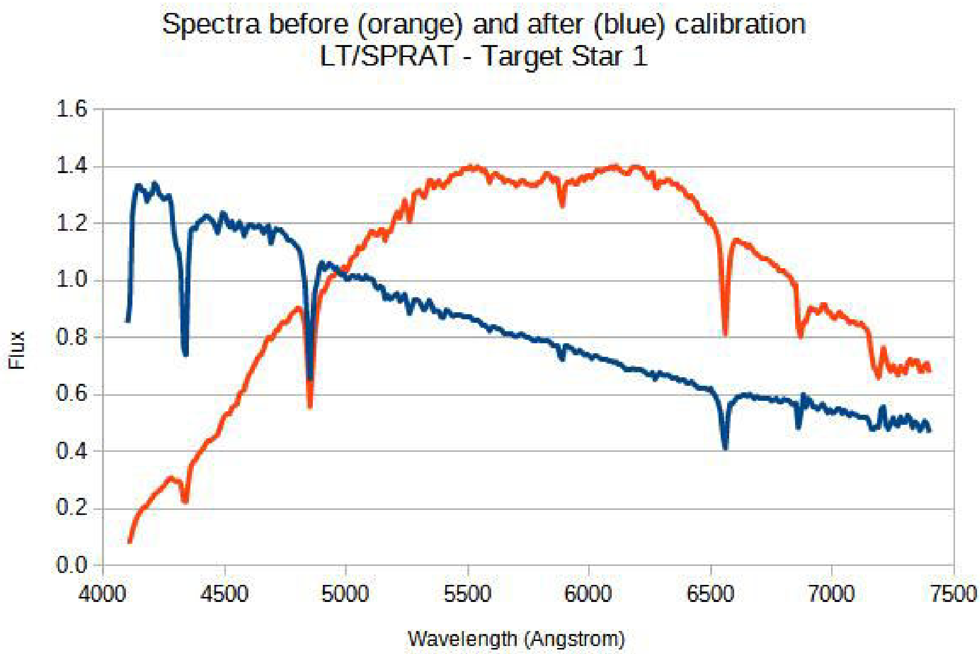

Figure 2: Target Star 1

- Star 2 - calibrated with Standard Calibration Star a

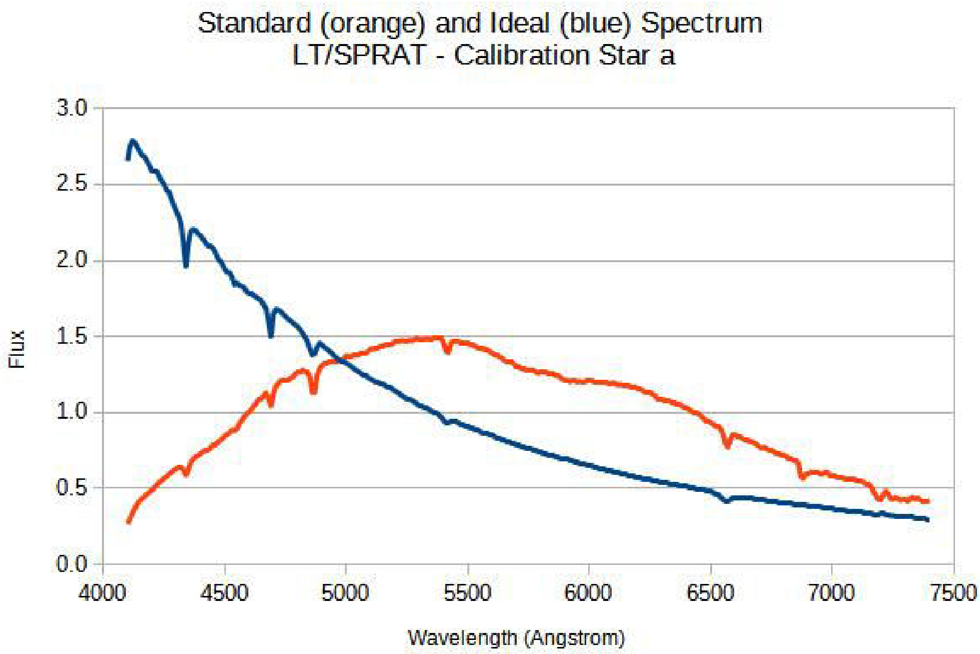

Figure 3: Calibration Star a

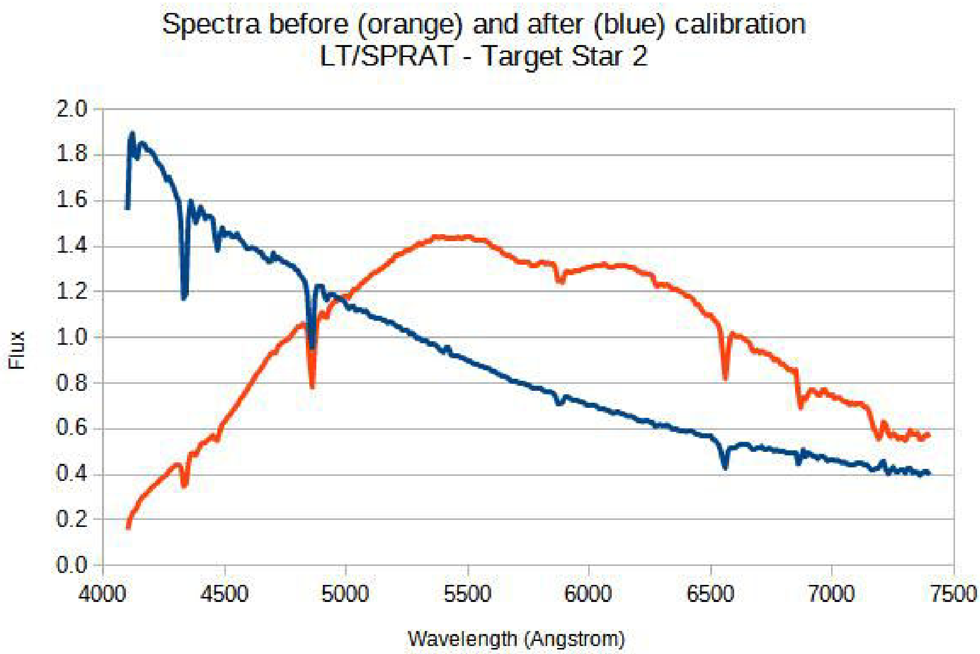

Figure 4: Target Star 2

- Star 3 - calibrated with Standard Calibration Star a

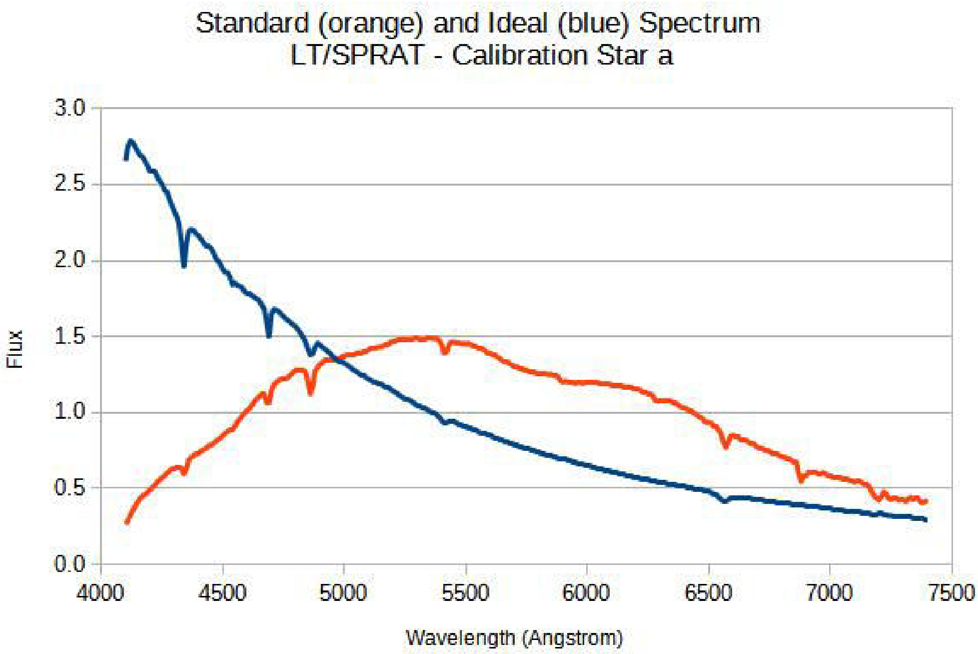

Figure 5: Calibration Star a

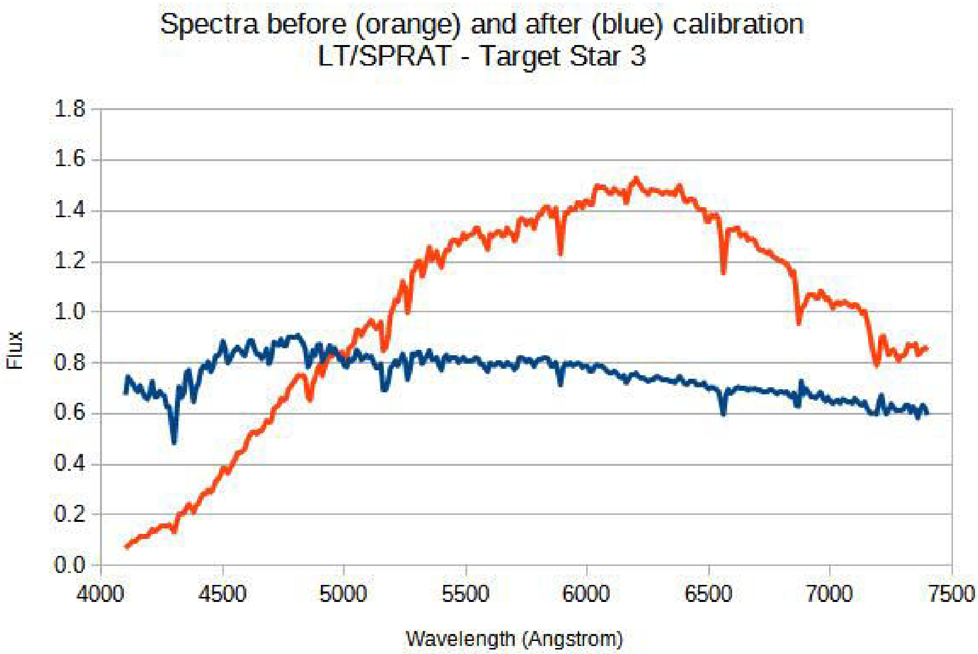

Figure 6: Target Star 3

• Star 4 - calibrated with Standard Calibration Star a

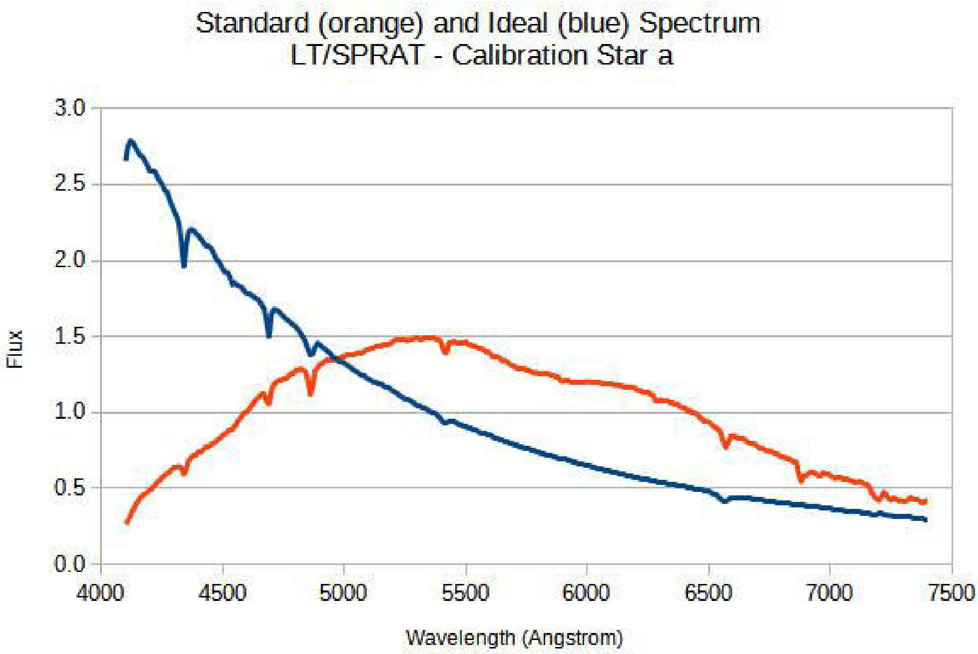

Figure 7: Calibration Star a

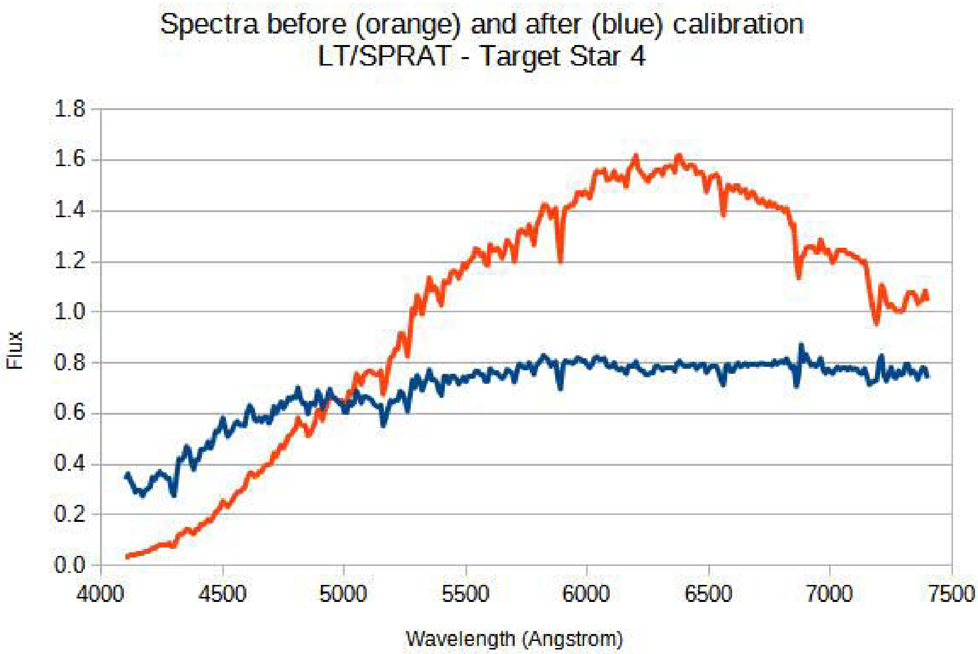

Figure 8: Target Star 4

• Star 5 - calibrated with Standard Calibration Star a

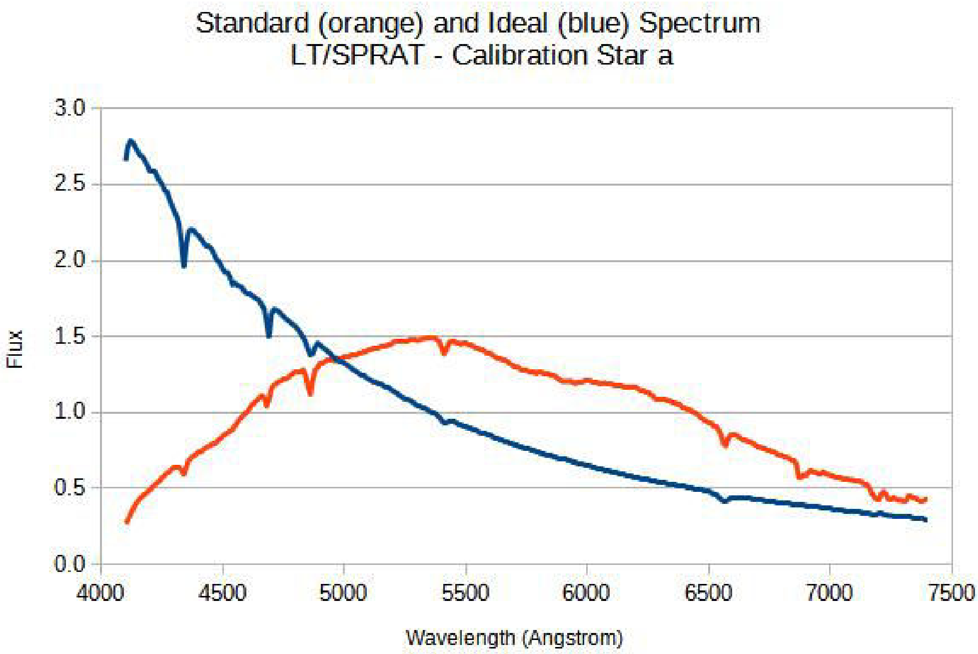

Figure 9: Calibration Star a

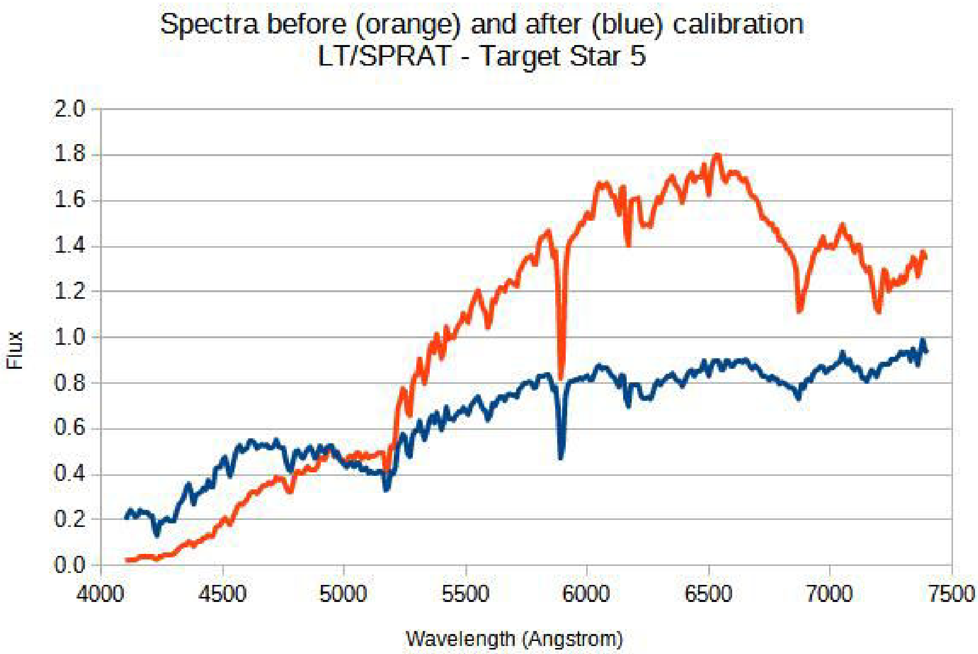

Figure 10: Target Star 5

- Star 6 - calibrated with Standard Calibration Star b

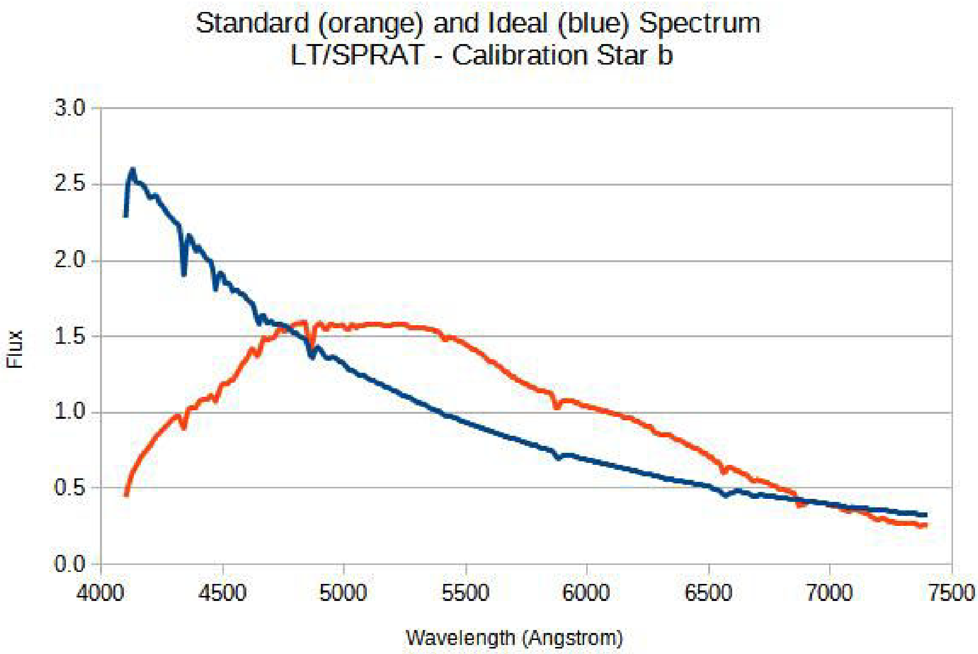

Figure 11: Calibration Star b

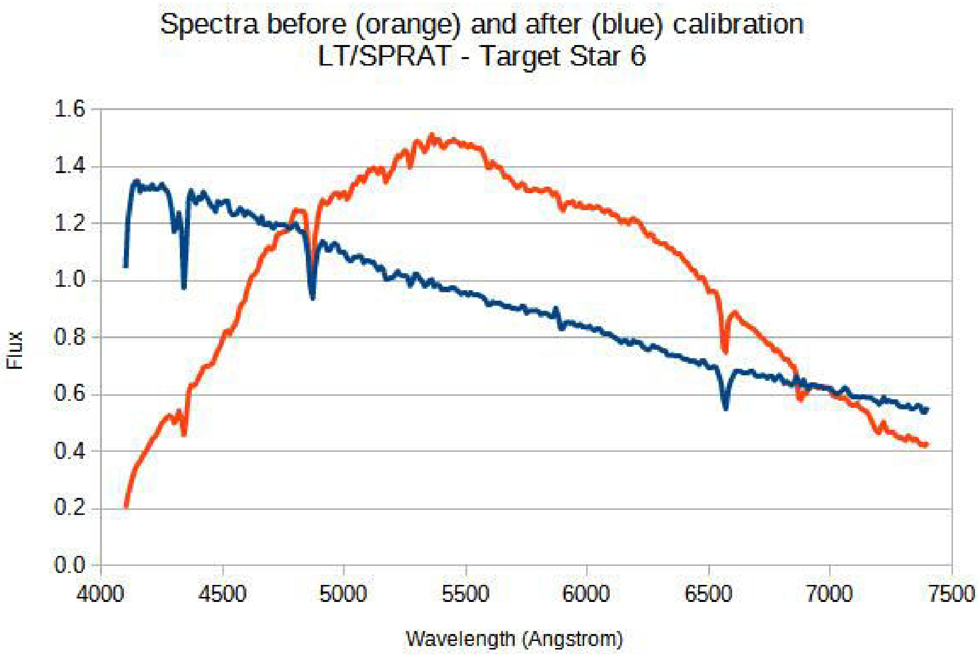

Figure 12: Target Star 6

- Star 7 - calibrated with Standard Calibration Star b

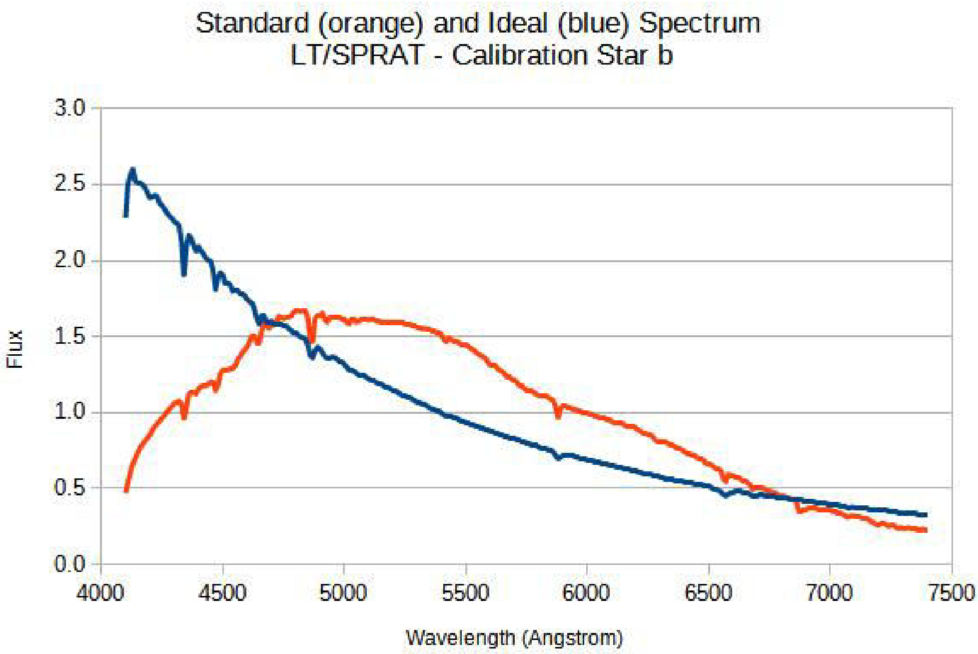

Figure 13: Calibration Star b

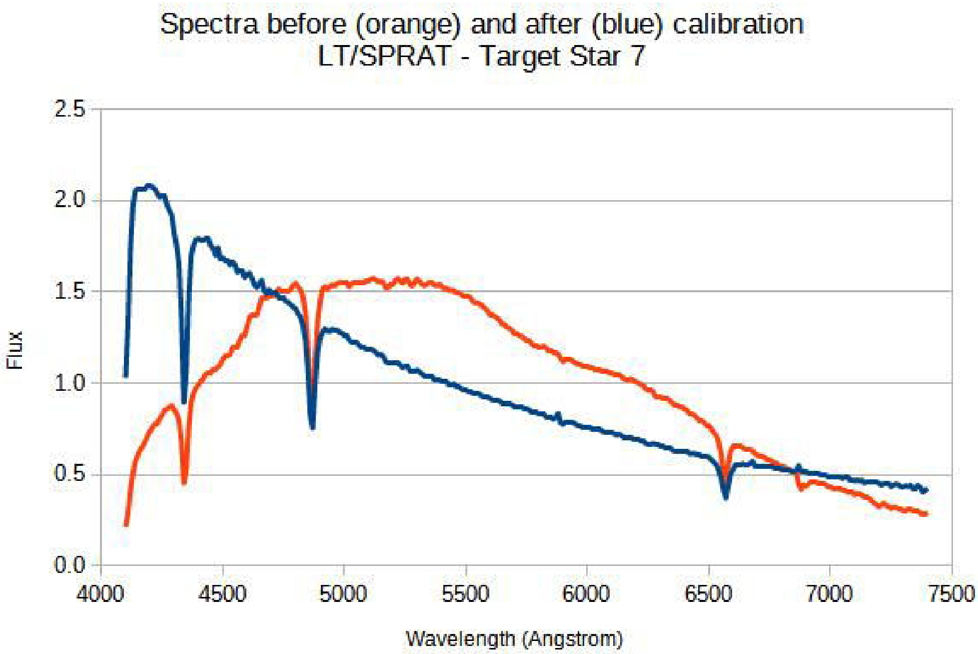

Figure 14: Target Star 7

- Star 8 - calibrated with Standard Calibration Star b

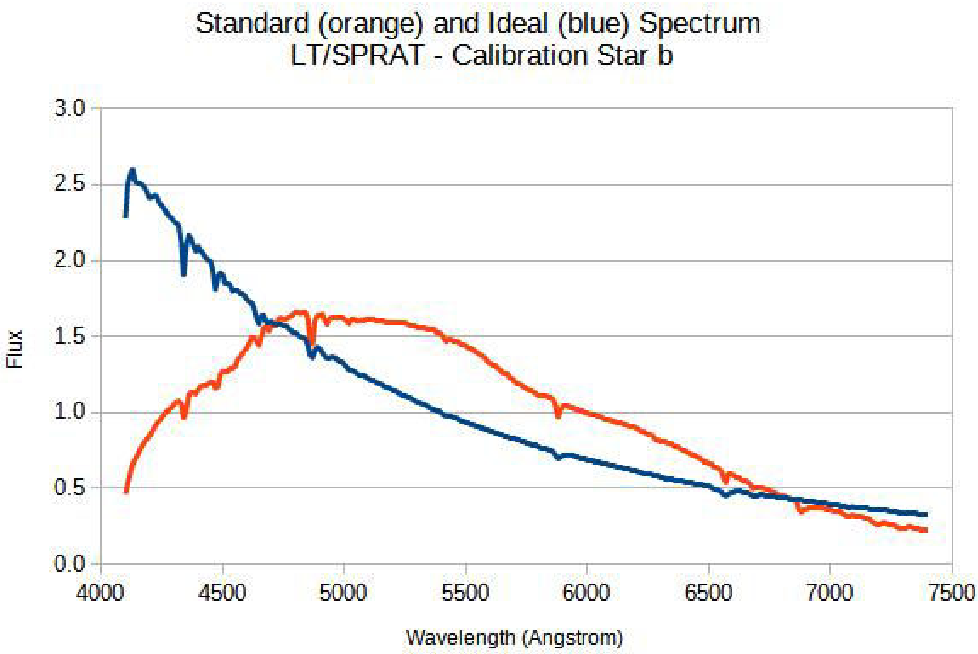

Figure 15: Calibration Star b

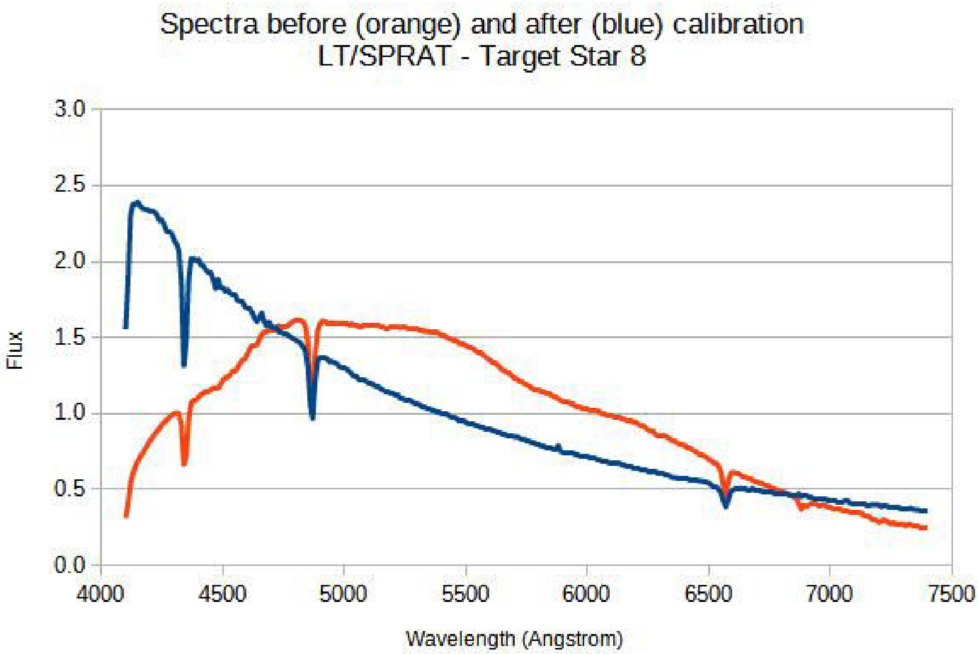

Figure 16: Target Star 8

- Star 9 - calibrated with Standard Calibration Star b

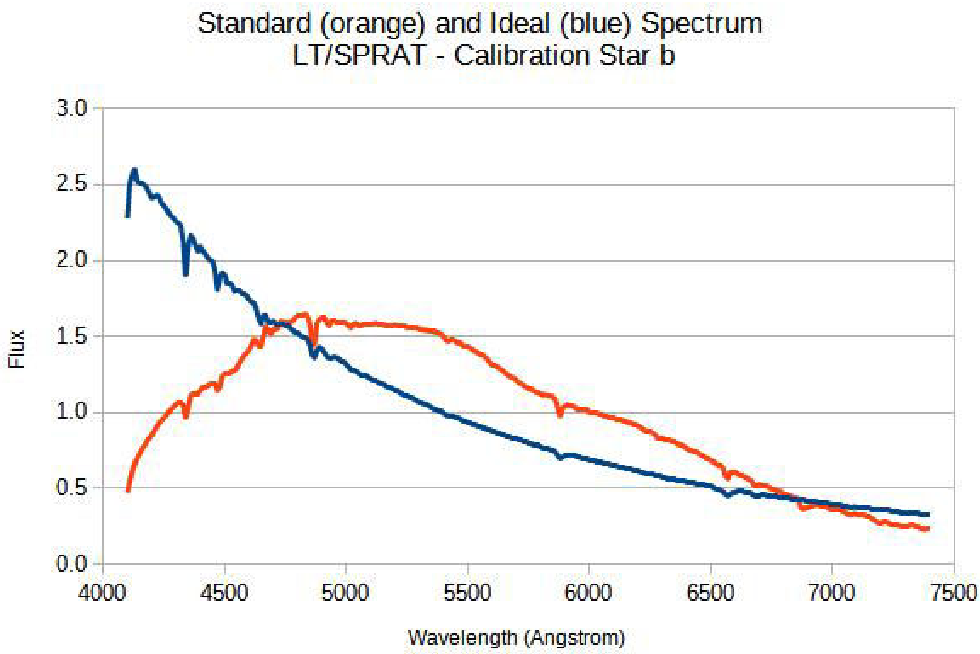

Figure 17: Calibration Star b

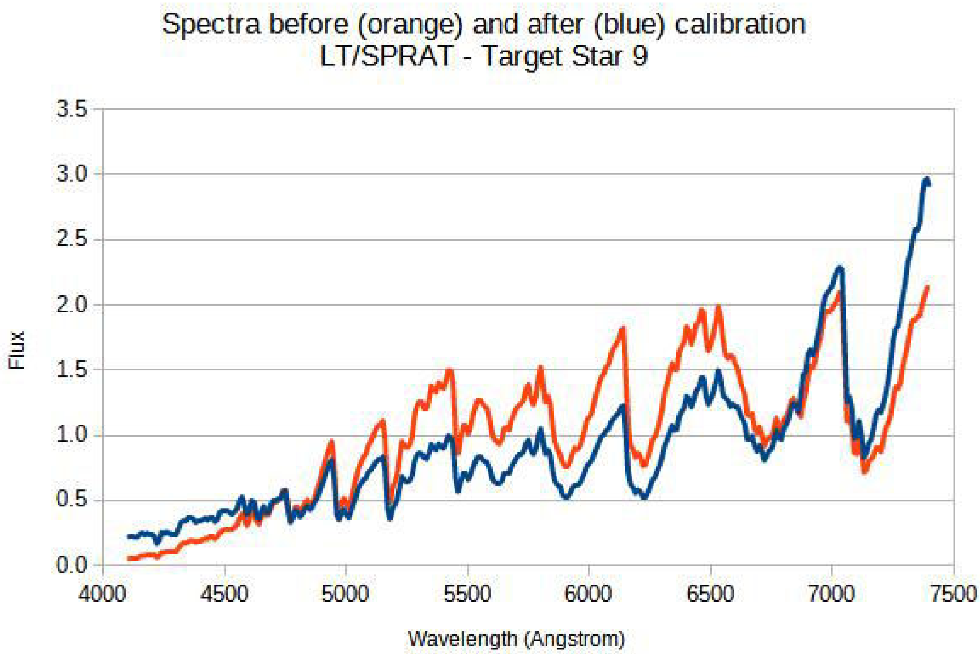

Figure 18: Target Star 9

3 Classification

Here each SPRAT Target Star is classified into different spectral type, according to the Spectral Classification Scheme (Spectral Class and sub-class only, without the Luminosity Class). There are 3 steps in the classification process:

1) Use the "least squares" method (analogous to the least squares method in statistical regression analysis) to select 3 spectra with the least squares differences between the calibrated flux and the Target Star and the template stars. It involves comparison of the strength of particular lines of the Target Stars and the template stars. This method is more systematic in finding the best fit for a set of data, as every single datum is taken into consideration and it is free from personal bias.

As an example, below is the calculation of the square difference between the calibrated flux of star 1 and HD 13505 for wavelength 4100 Angstrom:

Square of Difference = $(0.8505493 - 0.9397542)^2 = 0.0079575 \approx 0.01$.

Then sum the squares differences of all the wavelength is 3.32 for HD 13505S. See the extract of the spreadsheet in the following figure. The enclosed spreadsheet has more detailed calculations.

Wavelength (Angstrom)	Flux (Calibrated)	HD 16691	Square of difference	....	HD 13505	Square of difference	....	BD+361963	Square of difference
4100	0.8505493	2.5513160	2.89		0.9397542	0.01		0.2734788	0.33
4110	0.9159746	2.7917780	3.52		1.1161630	0.04		0.2610558	0.43
|	|	|	|		|	|		|	|
7390	0.4950492	0.2846225	0.04		0.5883094	0.01		6.1309340	31.76
7400	0.4659632	0.2835100	0.03	....	0.5932374	0.02	....	6.3679640	34.83
Sum of Square of difference		8.70			3.32			22.02	

Figure 19: Sum of Squares Differences - Star 1

2) For each of the selected spectra, calculate the λmax ("peak" brightness), given by Wien's Law, from the effective temperature Teff.

For example, HD 13505 is a F0 star with Teff of 7350K (refer to the table in figure 20), so its wavelength:

$$\lambda max = 2.898 \times 10^{-3} mK/7350K \times 10^{10} m = 3943\mathring{A}$$

3) Down-select one or more spectral class(s), which is the best fit spectrum, after considering the λmax, and other patterns of features, e.g. chemical compositions, absorption and emission lines, etc.. Figure 21 lists the Fraunhofer absorption lines for different elements. As some personal judgement is involved, so the reasoning of the selection is given.

Spectral type characteristics

Main sequence stars (V)

Spectral Type	Temperature (K)	Absolute Magnitude	Luminosity (in solar luminosities)
O5	54,000	-10.0	846,000
O6	45,000	-8.8	275,000
O7	43,300	-8.6	220,000
O8	40,600	-8.2	150,000
O9	37,800	-7.7	95,000
B0	29,200	-6.0	20,000
B1	23,000	-4.4	4600
B2	21,000	-3.8	2600
B3	17,600	-2.6	900
B5	15,200	-1.6	360
B6	14,300	-1.2	250
B7	13,500	-0.84	175
B8	12,300	-0.23	100
B9	11,400	0.29	62
A0	9600	1.4	22
A1	9330	1.6	18
A2	9040	1.8	15
A3	8750	2.1	12
A4	8480	2.3	10
A5	8310	2.4	9.0
A7	7920	2.7	6.7
F0	7350	3.2	4.3
F2	7050	3.5	3.3
F3	6850	3.7	2.8
F5	6700	3.8	2.4
F6	6550	4.0	2.1
F7	6400	4.1	1.8
F8	6300	4.2	1.7

Spectral Type	Temperature (K)	Absolute Magnitude	Luminosity (in solar luminosities)
G0	6050	4.5	1.3
G1	5930	4.6	1.2
G2	5800	4.8	1
G5	5660	4.9	0.86
G8	5440	5.2	0.68
K0	5240	5.4	0.54
K1	5110	5.6	0.46
K2	4960	5.8	0.38
K3	4800	6.0	0.31
K4	4600	6.3	0.24
K5	4400	6.6	0.19
K7	4000	7.3	0.10
M0	3750	7.7	0.069
M1	3700	7.8	0.064
M2	3600	7.9	0.054
M3	3500	8.1	0.046
M4	3400	8.3	0.038
M5	3200	8.7	0.026
M6	3100	8.9	0.022
M7	2900	9.4	0.014
M8	2700	9.9	0.0093
L0	2600	*	0.0074
L3	2200	*	0.0027
L8	1500	*	0.00026
T2	1400	*	0.00017
T6	1000	*	0.000021
T8	800	*	0.0000055

*- not visible to the human eye (for the most part)

Figure 20: source: https://sites.uni.edu/morgans/astro/course/Notes/section2/spectraltemps.html

Designation	Element	Wavelength (nm)	Designation	Element	Wavelength (nm)
y	O_2	898.765	c	Fe	495.761
Z	O_2	822.696	F (Hβ)	H	486.134
A	O_2	759.370	d	Fe	466.814
B	O_2	686.719	e	Fe	438.355
C (Hα)	H	656.281	G' (Hγ)	H	434.047
a	O_2	627.661	G	Fe	430.790
D_1	Na	589.592	G	Ca	430.774
D_2	Na	588.995	h (Hδ)	H	410.175
D_3 or d	He	587.5618	H	Ca^+	396.847
e	Hg	546.073	K	Ca^+	393.368
E_2	Fe	527.039	L	Fe	382.044
b_1	Mg	518.362	N	Fe	358.121
b_2	Mg	517.270	P	Ti^+	336.112
b_3	Fe	516.891	T	Fe	302.108
b_4	Mg	516.733	t	Ni	299.444

Figure 21: Fraunhofer Lines, source: https://en.m.wikipedia.org/wiki/Astronomical$_{s}pectroscopy$

- Star 1

The table below lists the 3 template stars with the "least squares differences" when comparing with the Target Star, the corresponding spectral classes, temperature, and λmax.

Template Star	Class	Temp.	$\lambda max \mathring{A}$
HD 13505	F0	7350	3943
HZ 948	F3	6850	4231
HD 83140	F3	6850	4231

F stars are white stars, with temperature between 6,000-7,000K. Their chemical components include singly ionised Ca (CaII), neutral H weaker, and other metals. Star 1 is more likely to be a F3 star, as its λmax is higher than 4000 Å, which is a cooler star. Its absorption lines for neutral H is quite obvious at 6563Å, 4861Å and 4340Å.

The gaps in the comparisons may be due to the noise, as the template spectra are of real stars and so will not all be "ideal".

See the figures below for their comparisons.

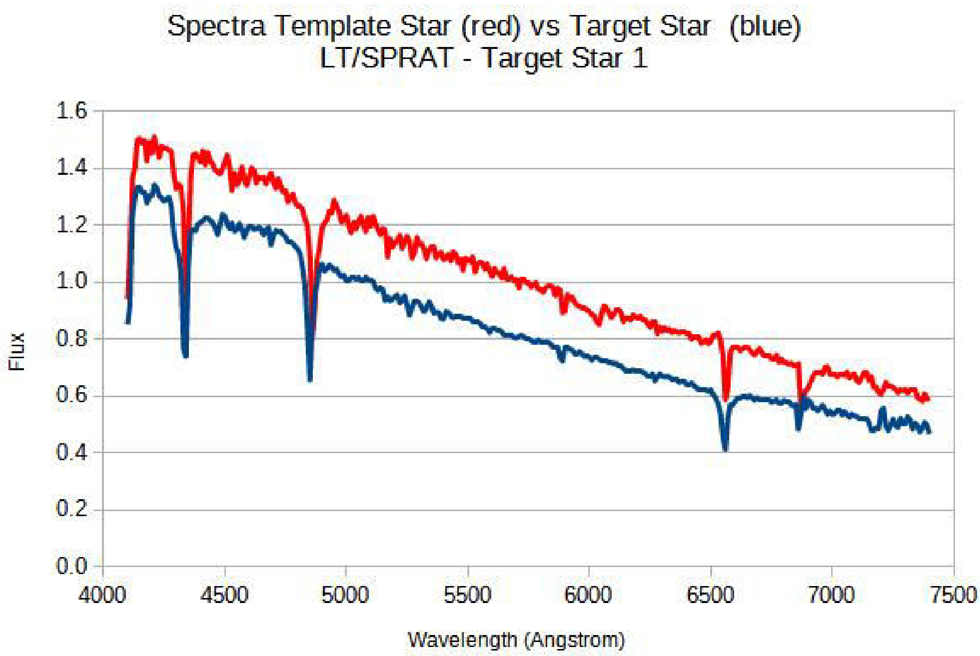

Figure 22: Star 1 vs HD 13505

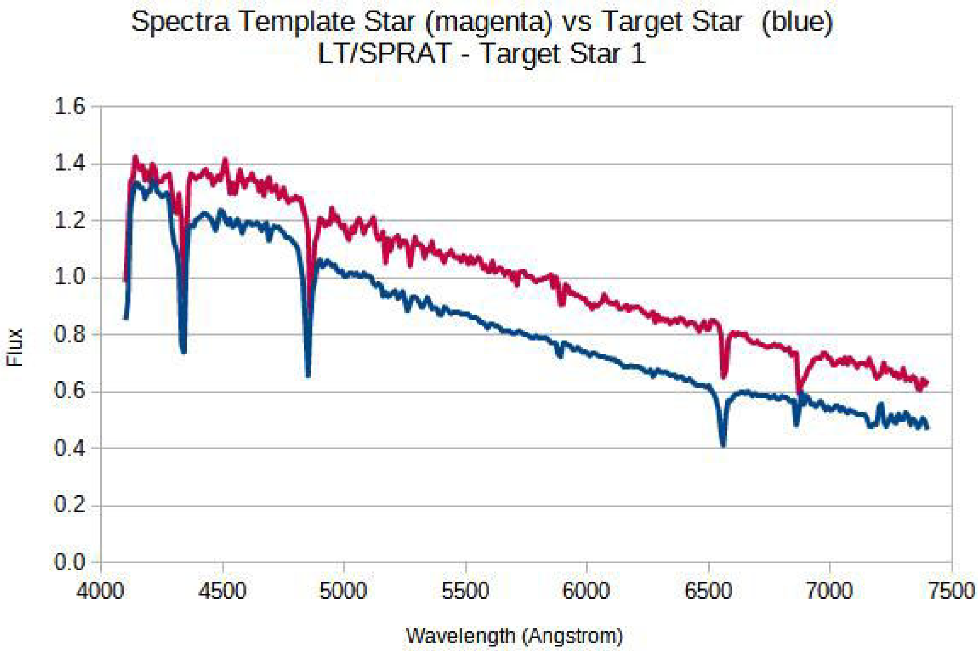

Figure 23: Star 1 vs HZ 948

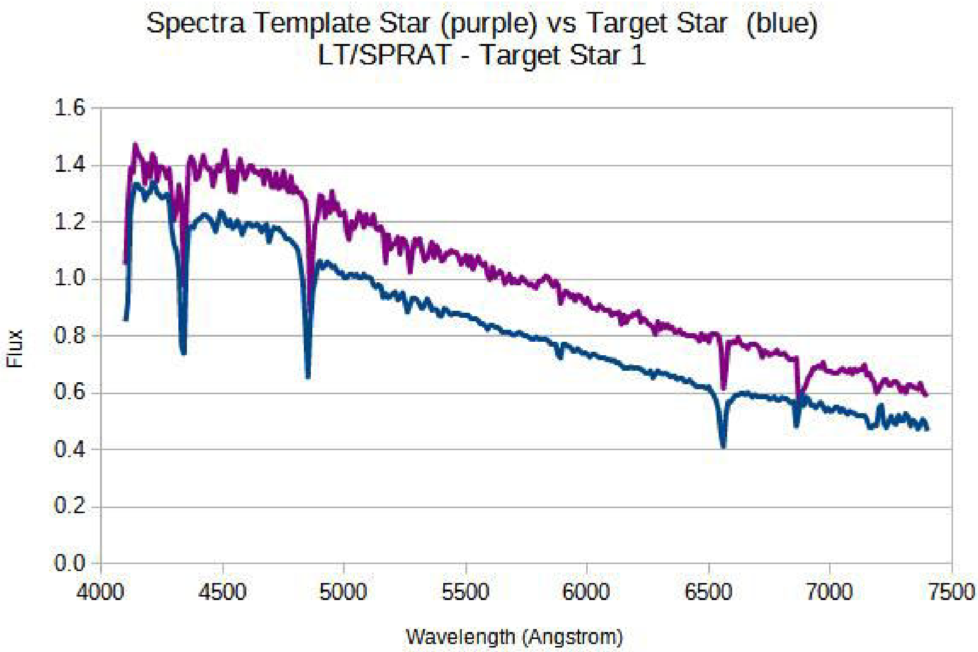

Figure 24: Star 1 vs HD 83140

- Star 2

The table below lists the 3 template stars with the "least squares differences" when comparing with the Target Star, the corresponding spectral classes, temperature, and λmax.

Template Star	Class	Temp.	$\lambda max \mathring{A}$
SAO 87116	A3	8750	3312
SAO 12149	A1	9330	3106
SAO 37370	F0	7350	3943

A stars are blue/blue-white stars, neutral H (Balmer lines) dominate, with temperature between 7,000-10,000K. F stars are white stars, with temperature between 6,000-7,000K. Their chemical components include singly ionised Ca (CaII), neutral H weaker, and other metals. Star 2 is more likely to be an A3 star, as its absorption lines have better matching with the Target Star.

See the figures below for their comparisons.

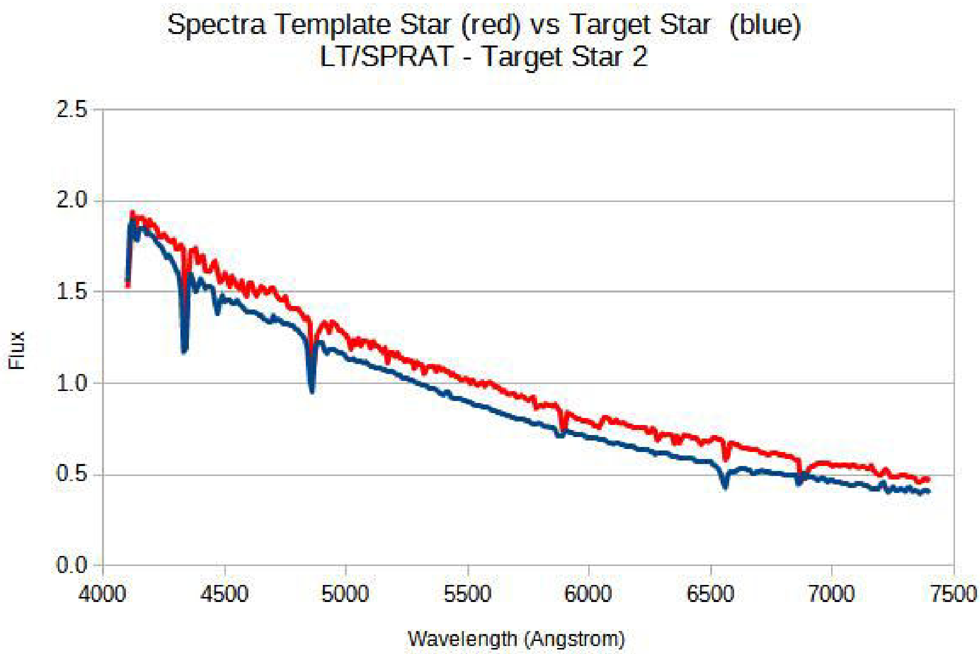

Figure 25: Star 2 vs SAO 87716

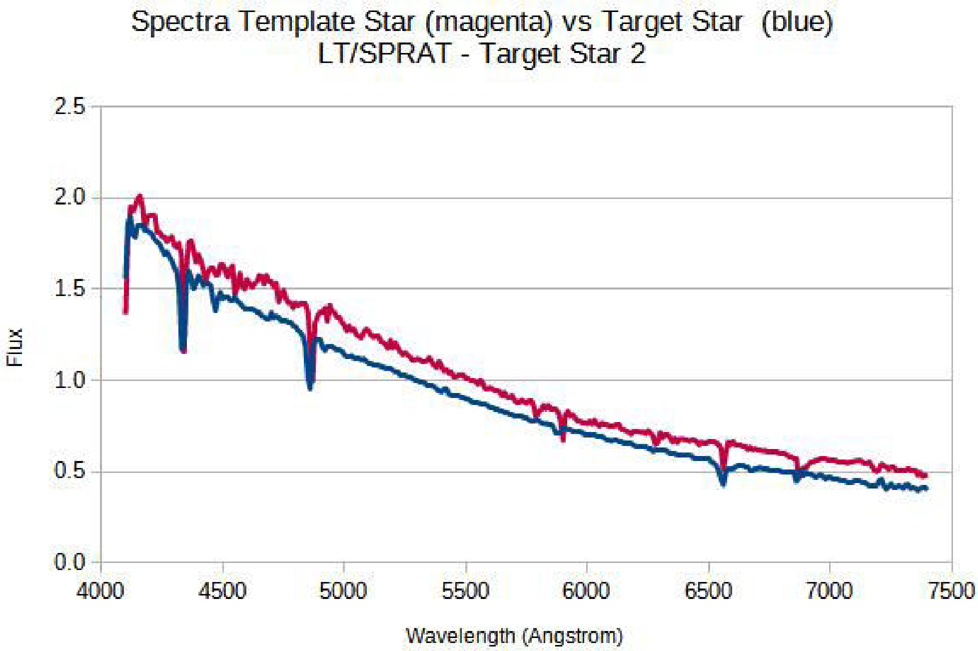

Figure 26: Star 2 vs SAO 12149

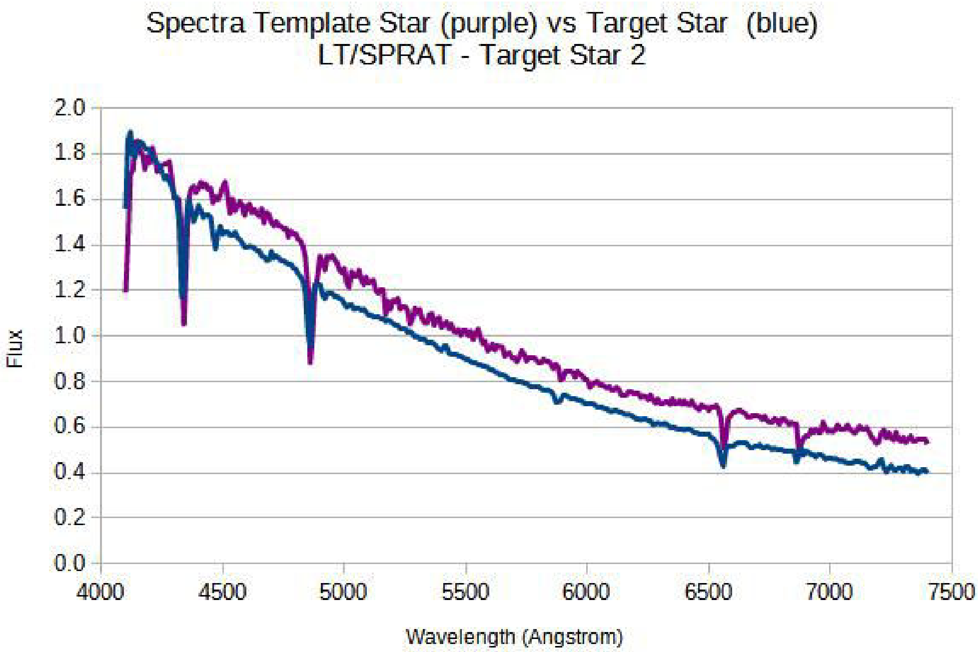

Figure 27: Star 2 vs SAO 37370

Star 3

The table below lists the 3 template stars with the "least squares differences" when comparing with the Target Star, the corresponding spectral classes, temperature, and λmax.

Template Star	Class	Temp.	$\lambda max \AA$
HD 25894	G2	5800	3943
HD 23524	K0	5240	5531
TRA 14	G4	5700*	5084

* Estimated by interpolation between temperature of G2 and G5 stars.

G stars are yellow stars, Ca II dominates, neutral metals (e.g. FeI) with temperature between 5,000-6,000K. K stars are orange-red stars, with neutral metals (Ca, Fe) dominate, molecular bands appear with temperature between 3,500-5,000K. Star 3 can be a G4 star, as its λmax is at around 5000 Å. However, there are some uncertainties here as the absorption lines of Ca, and Fe, which are below 4000Å, are not available.

See the figures below for their comparisons.

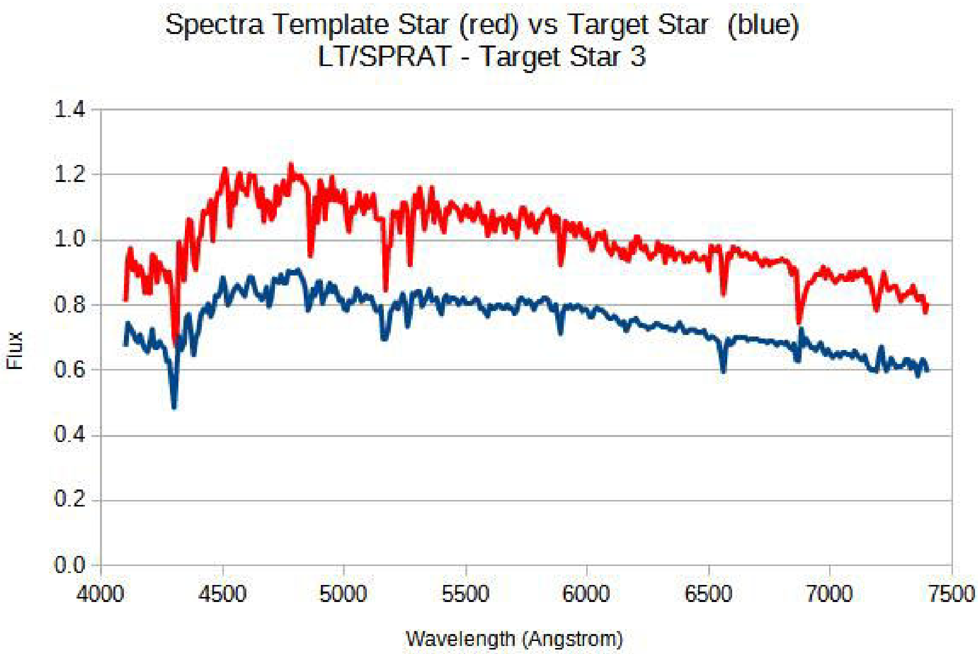

Figure 28: Star 3 vs HD 25894

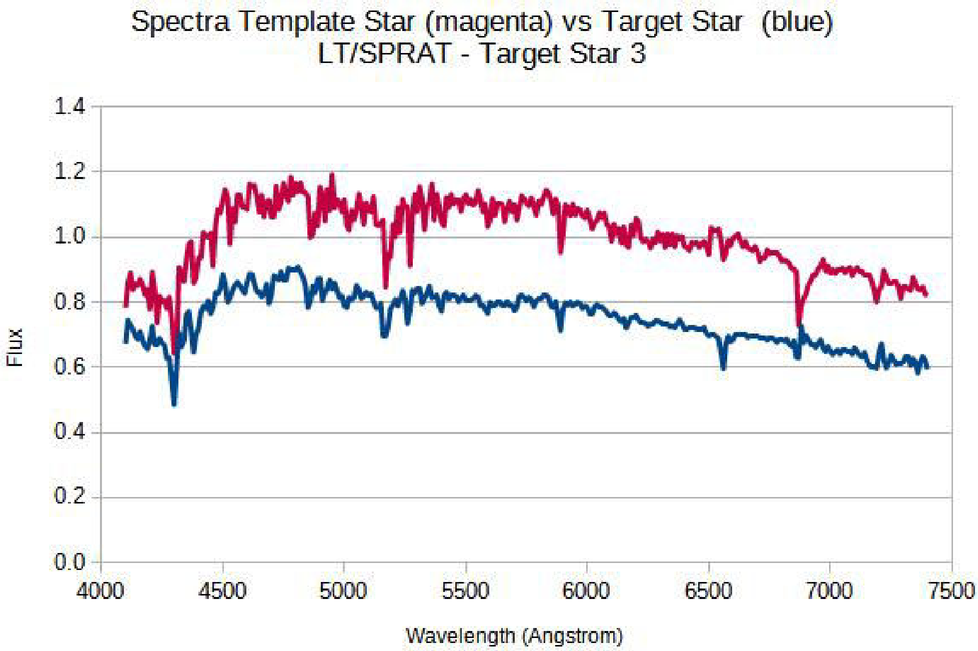

Figure 29: Star 3 vs HD 23524

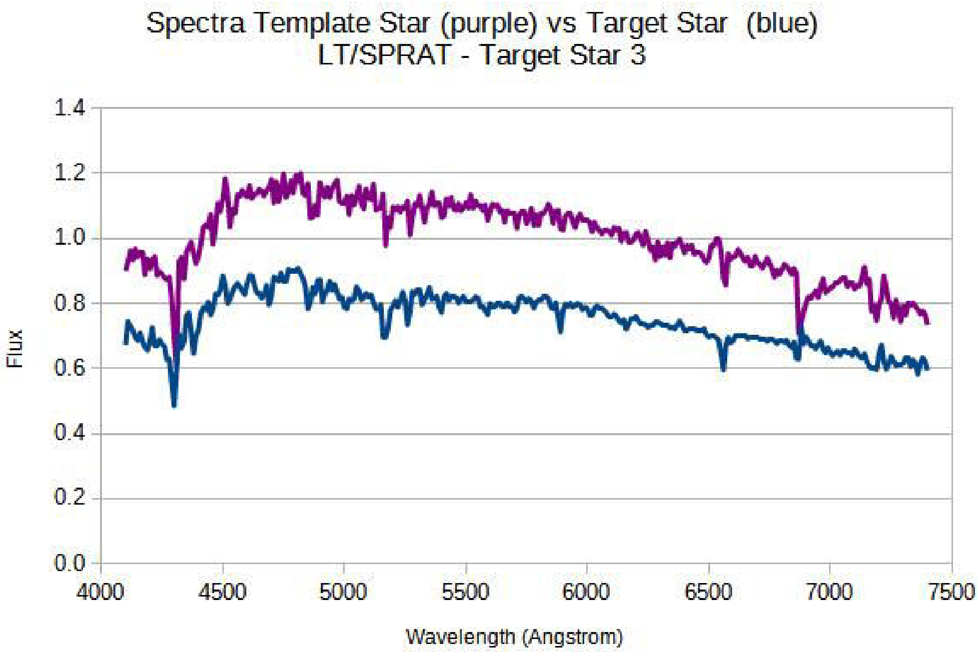

Figure 30: Star 3 vs TRA 14

- Star 4

The table below lists the 3 template stars with the "least squares differences" when comparing with the Target Star, the corresponding spectral classes, temperature, and λmax.

Template Star	Class	Temp.	$\lambda max \mathring{A}$
HD 249384	G8	5440	5327
HD 250368	G9	5340*	5427
SAO 55164	K0	5240	5531

* Estimated by interpolation between temperature of G8 and K0 stars.

G stars are yellow stars, Ca II dominates, neutral metals (e.g. FeI) with temperature between 5,000-6,000K. K stars are orange-red stars, with neutral metals (Ca, Fe) dominate, molecular bands appear with temperature between 3,500-5,000K. Star 4 can be K0, as its λmax is at higher than 5500 Å. Again, uncertainties exist, as the absorption lines of Ca, and Fe, which are below 4000Å, are not available.

See the figures below for their comparisons.

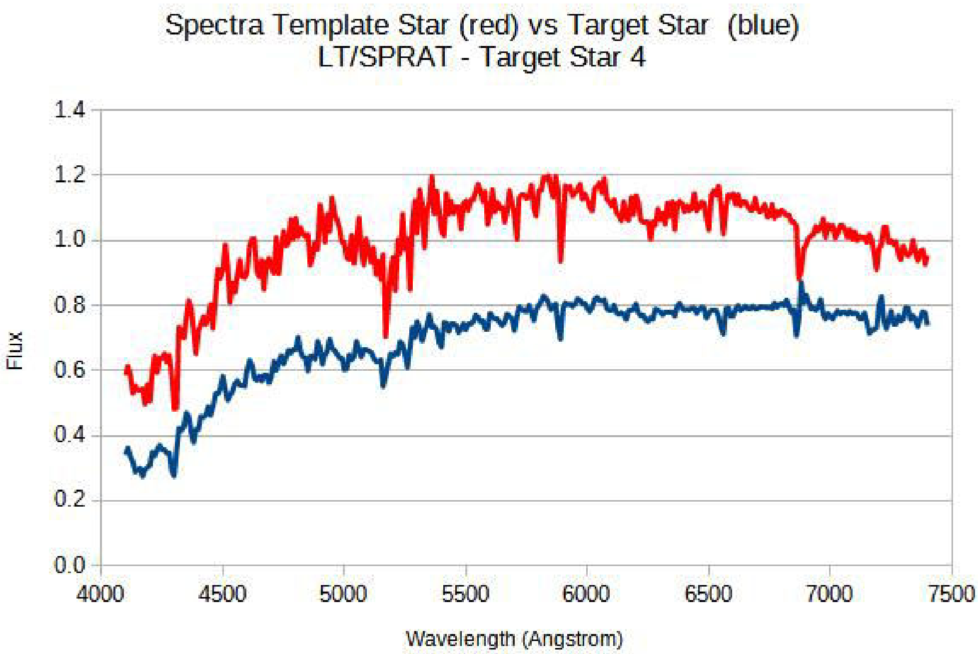

Figure 31: Star 4 vs HD 249384

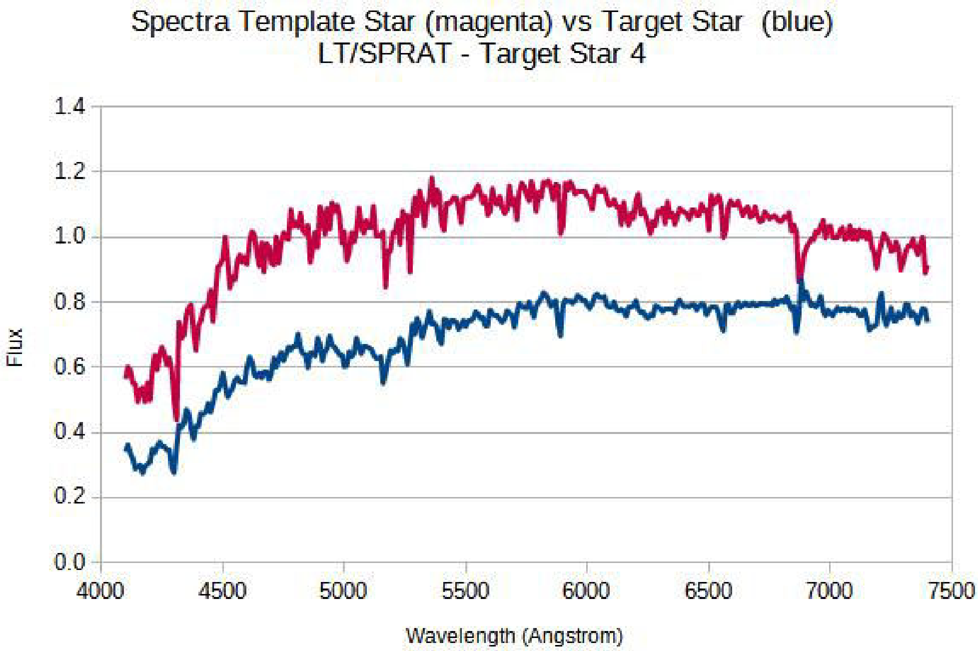

Figure 32: Star 4 vs HD 250368

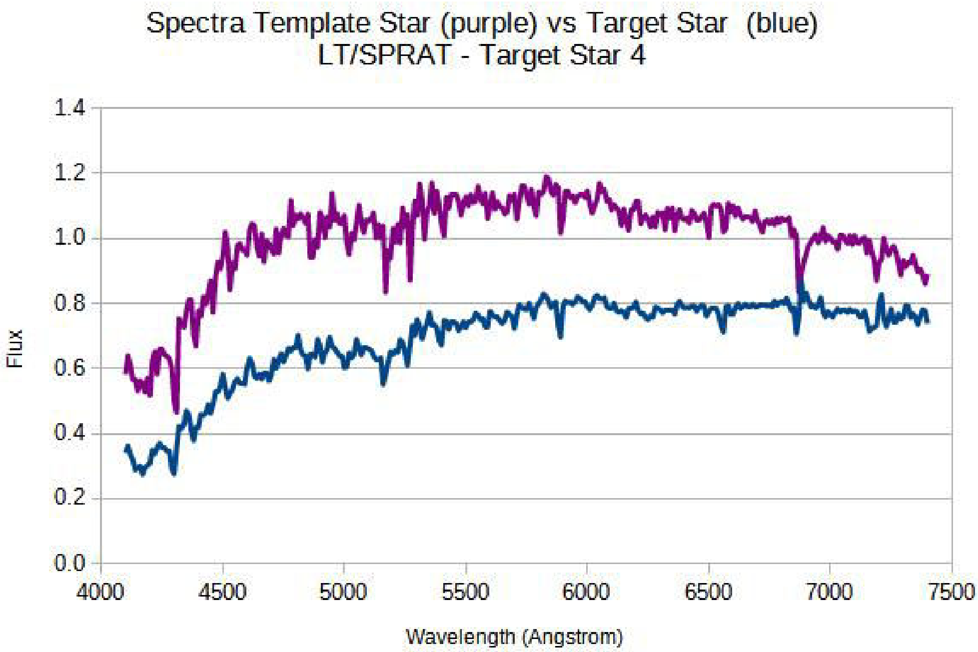

Figure 33: Star4 vs SAO55164

- Star 5

The table below lists the 3 template stars with the "least squares differences" when comparing with the Target Star, the corresponding spectral classes, temperature, and λmax.

Template Star	Class	Temp.	$\lambda max \mathring{A}$
HD 26946	K3	4800	6038
SAO 76803	K5	4400	6586
HD 5351	K4	4600	6300

K stars are orange-red stars, with neutral metals (Ca, Fe) dominate, molecular bands appear with temperature between 3,500-5,000K. Star 5 is likely a K5 star, as its λmax is close to the 7000 Å. Again, uncertainties exist, as the absorption lines of Ca, and Fe, which are below 4000Å, are not available.

See the figures below for their comparisons.

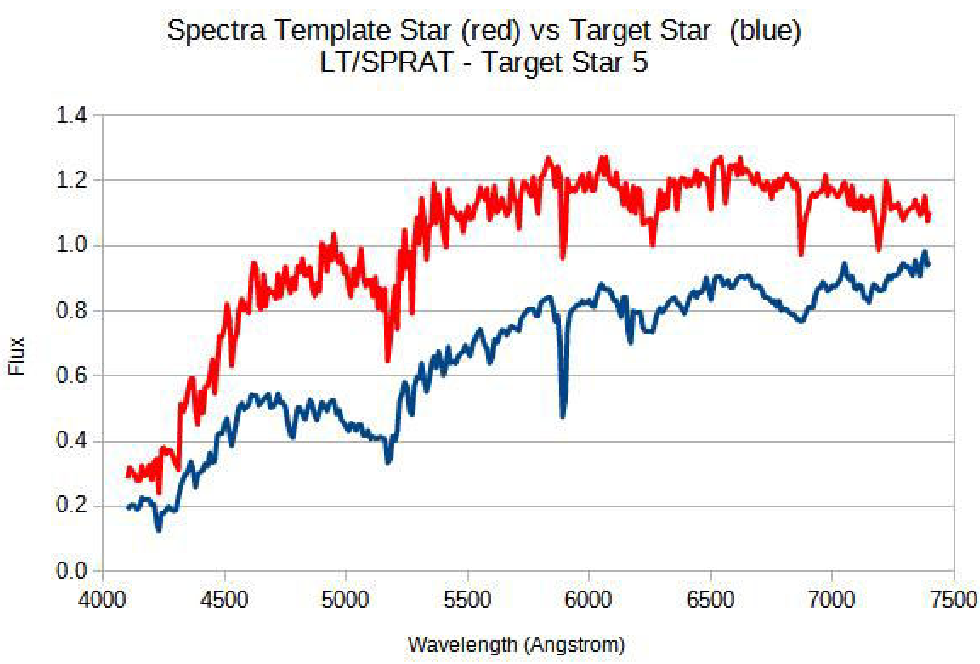

Figure 34: Star 5 vs HD 26946

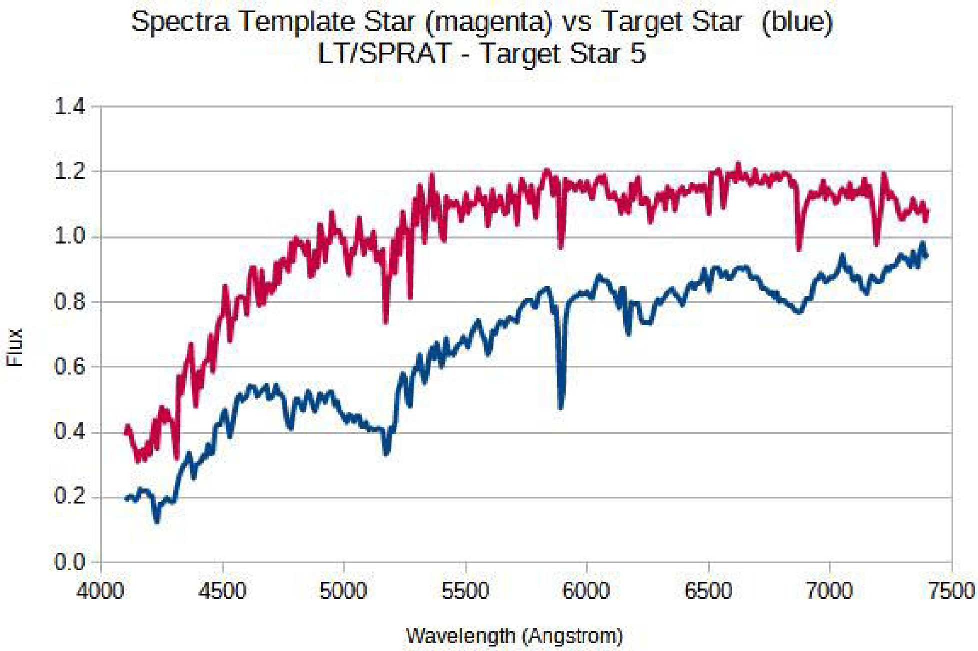

Figure 35: Star 5 vs SAO 76803

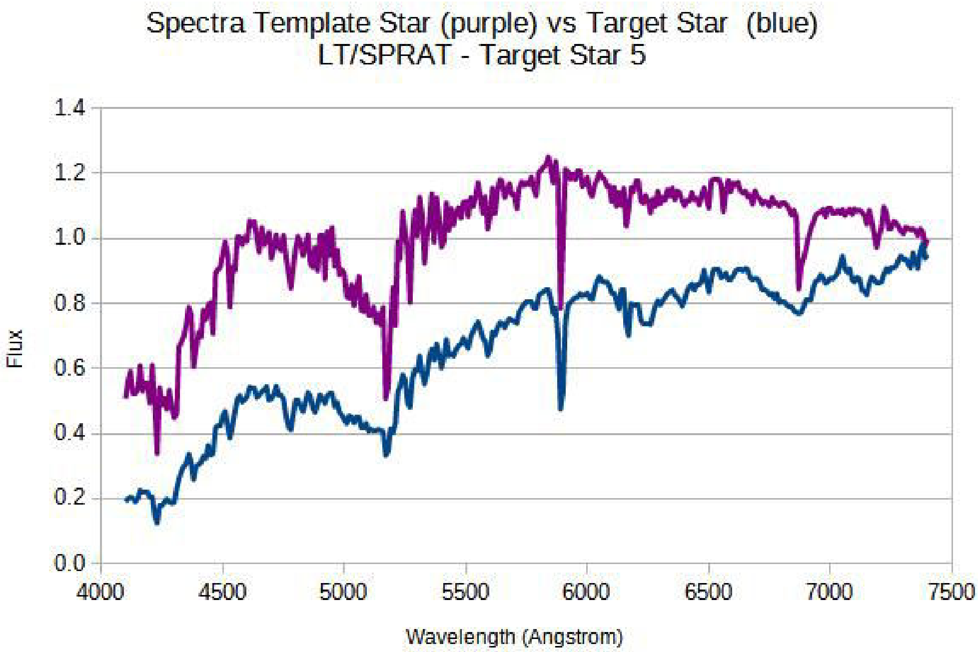

Figure 36: Star 5 vs HD 5351

- Star 6

The table below lists the 3 template stars with the "least squares differences" when comparing with the Target Star, the corresponding spectral classes, temperature, and λmax.

Template Star	Class	Temp.	$\lambda max \mathring{A}$
HZ 948	F3	6850	4231
HD 83140	F3	6850	4231
BD+61 367	F5	6700	4325

F stars are white stars, with temperature between 6,000-7,000K, and have chemical components of singly ionised Ca (CaII), neutral H weaker, and other metals. Star 6 can be either a F3 or F5 star, as its absorption lines of the elements all appear in the spectra.

See the figures below for their comparisons.

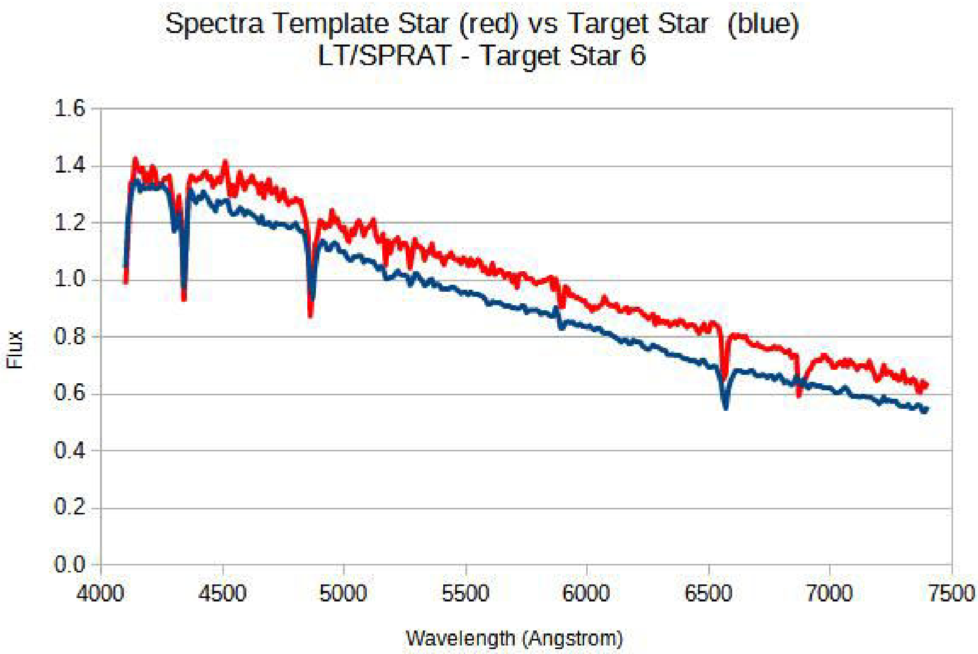

Figure 37: Star 6 vs HZ 948

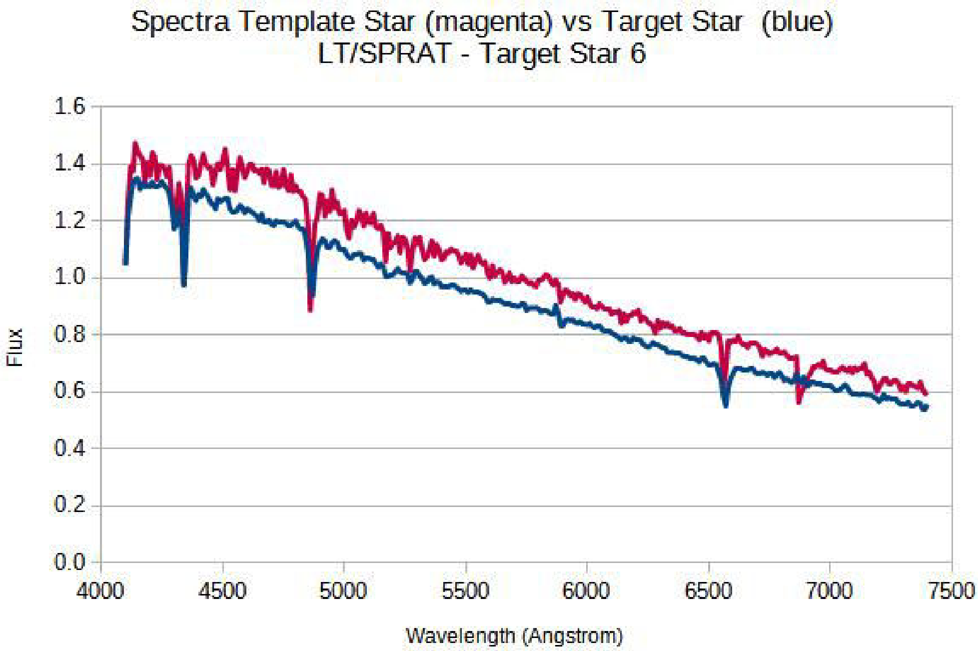

Figure 38: Star 6 vs HD 83140

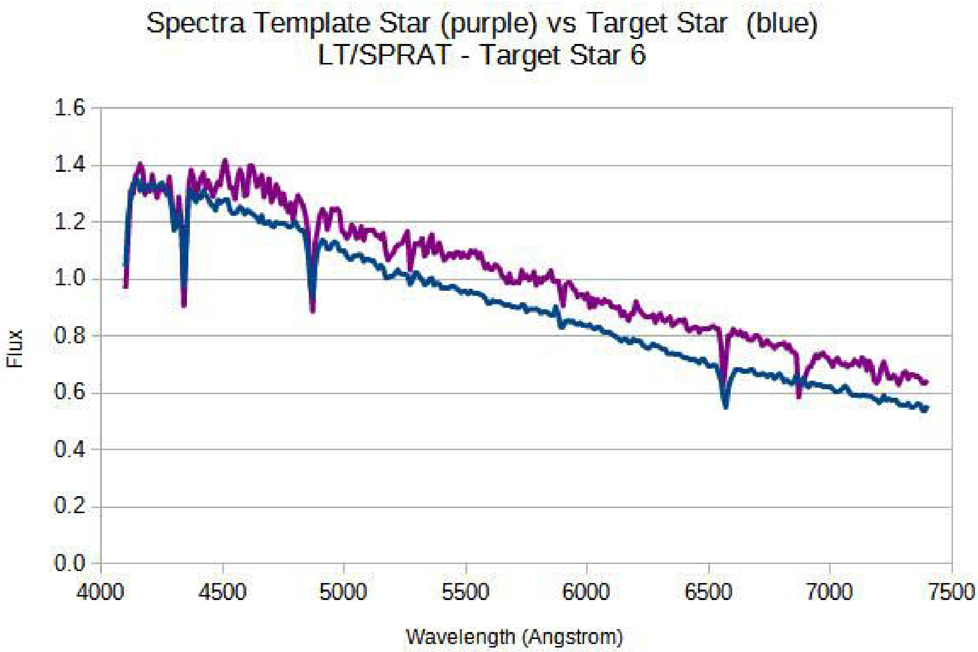

Figure 39: Star 6 vs BD+61 367

- Star 7

The table below lists the 3 template stars with the "least squares differences" when comparing with the Target Star, the corresponding spectral classes, temperature, and λmax.

Template Star	Class	Temp.	$\lambda max \mathring{A}$
HD 190785	A2	9040	3206
HD 12027	A3	8750	3312
HD 221741	A3	8750	3312

A stars are blue/blue-white stars, with neutral H (Balmer lines) dominate, and with temperature between 7,000-10,000K. Star 7 can be either a A2 or A3 star, or something else as its λmax is below 4000 Å.

See the figures below for their comparisons.

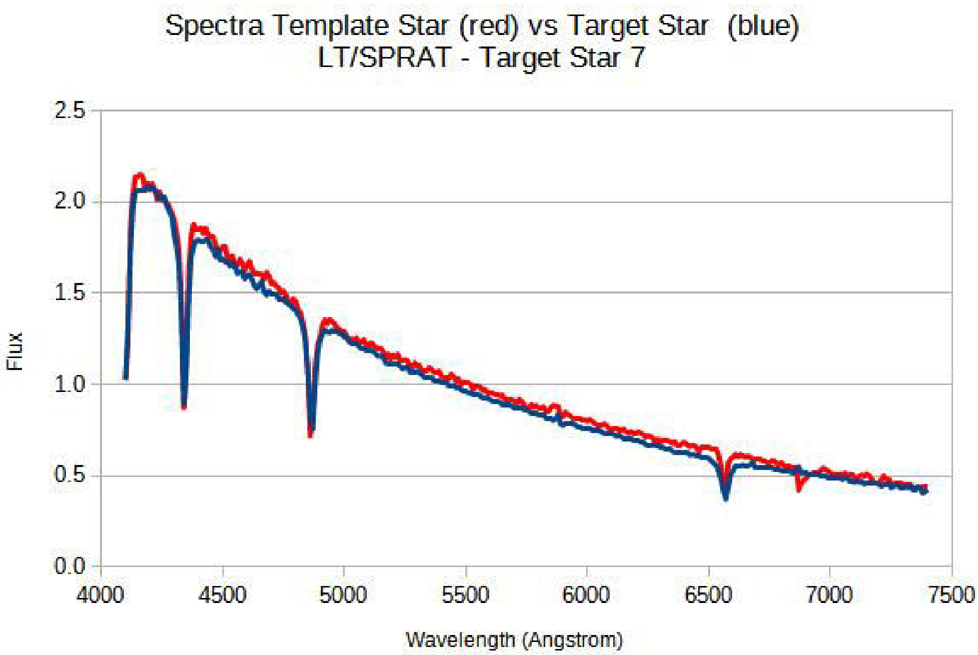

Figure 40: Star 7 vs HD 190785

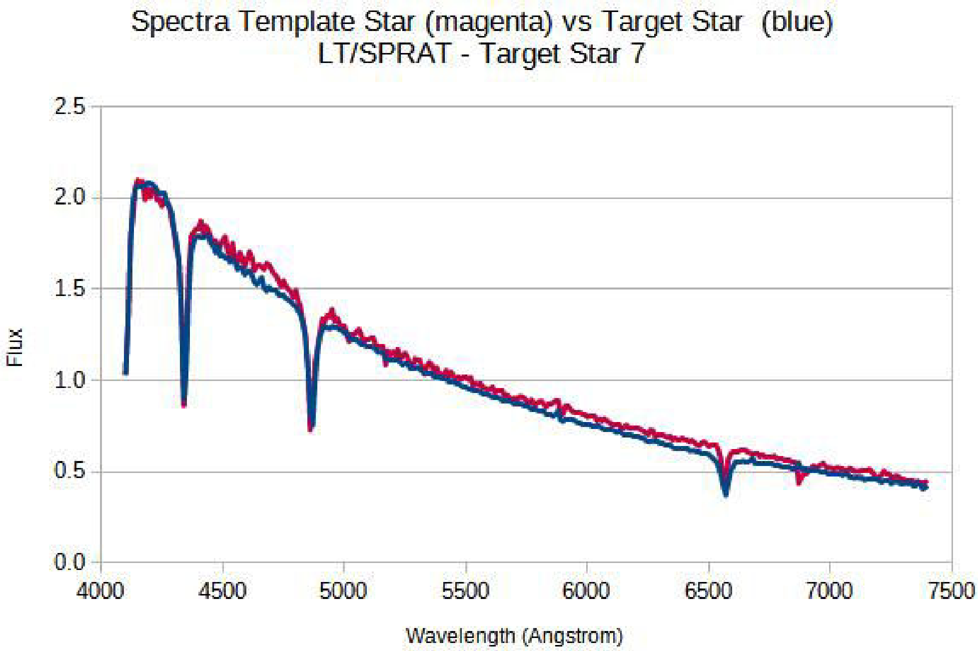

Figure 41: Star 7 vs HD 12027

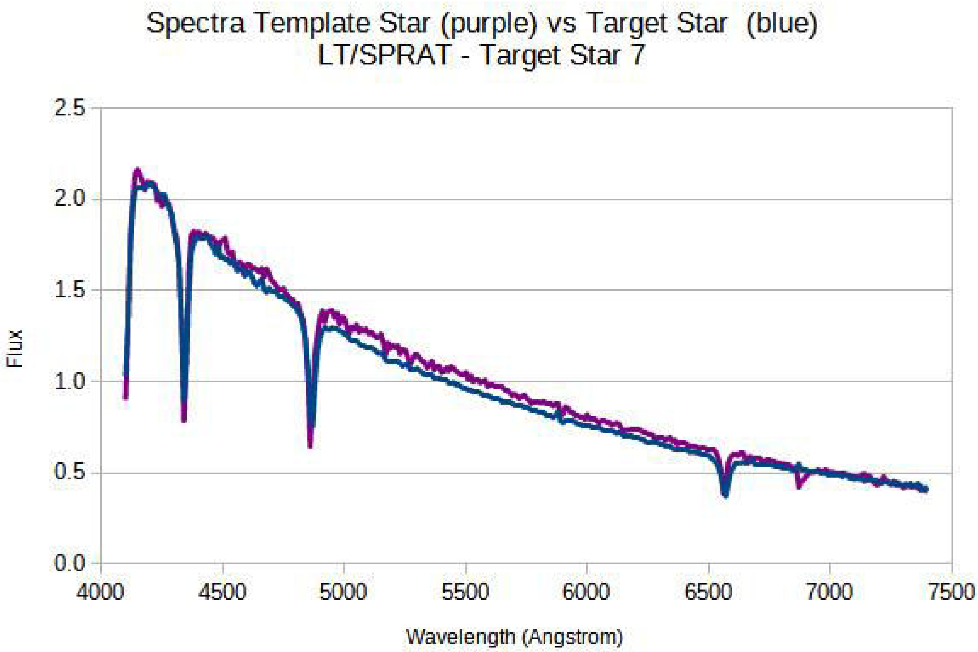

Figure 42: Star 7 vs HD 221741

- Star 8

The table below lists the 3 template stars with the "least squares differences" when comparing with the Target Star, the corresponding spectral classes, temperature, and λmax.

Template Star	Class	Temp.	$\lambda max \mathring{A}$
O 1015	B8	12300	2356
FEIGE 40	B4	16400*	1767
HD 30584	B6	14300	2027

* Estimated by interpolation between temperature of B3 and B5 stars.

B stars are hot blue stars, neutral He dominate (HeI), with temperature between 10,000-20,000K. Star 8 can be either a B4, B6 or B8 star, based on its overall absorption lines pattern. However, there is still some uncertainties, as the absorption line of He, at around 5876 Å, is not so obvious.

See the figures below for their comparisons.

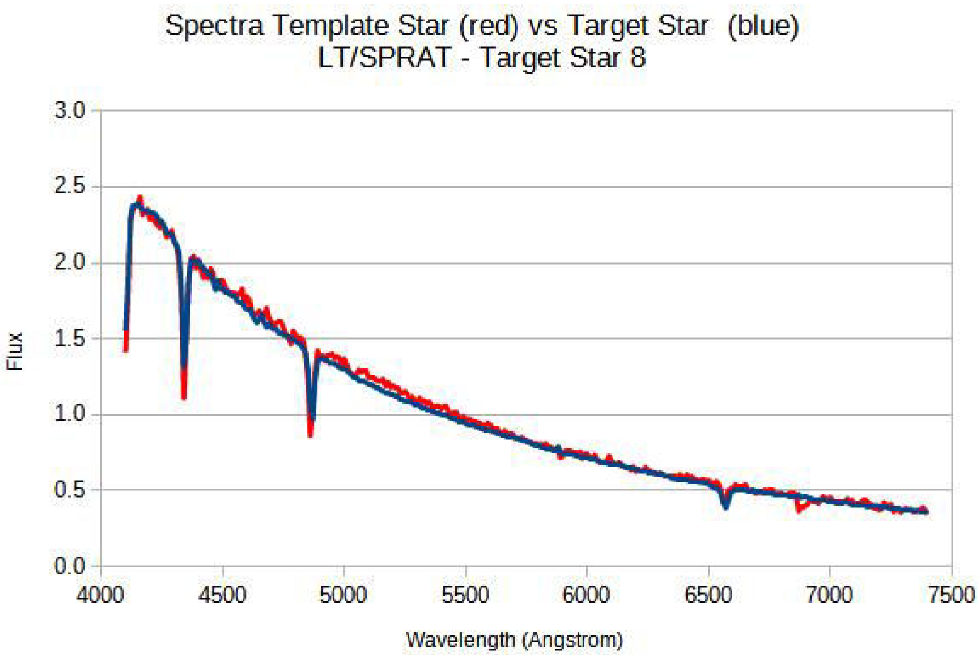

Figure 43: Star 8 vs O 1015

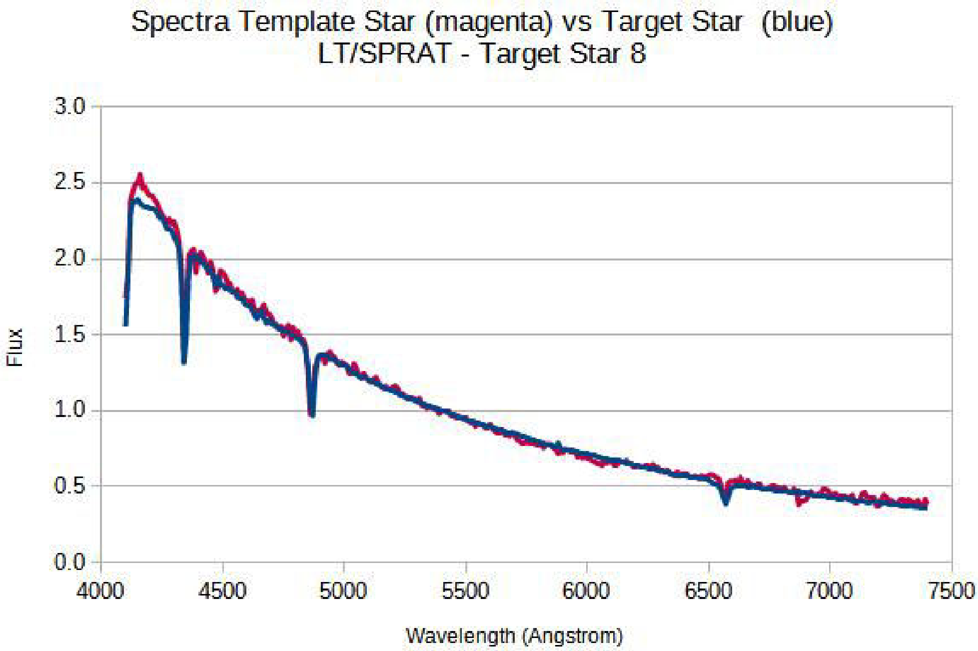

Figure 44: Star 8 vs FEIGE 40

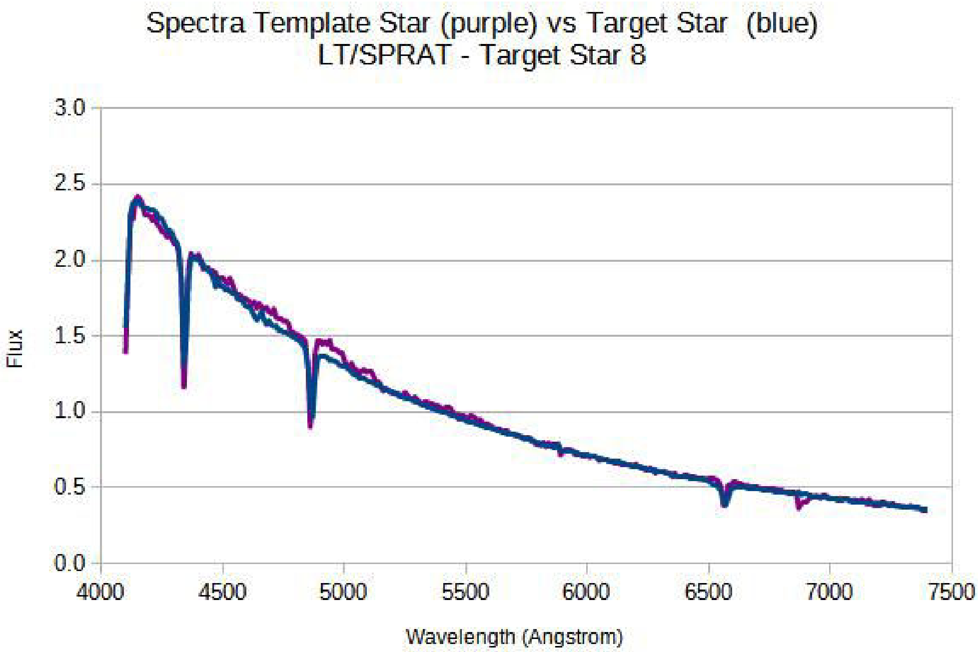

Figure 45: Star 8 vs HD 30584

- Star 9

The table below lists the 3 template stars with the "least squares differences" when comparing with the Target Star, the corresponding spectral classes, temperature, and λmax.

Template Star	Class	Temp.	$\lambda max \mathring{A}$
HD 110964	M4	3400	8524
SAO 63349	M3	3500	8280
HD 249826	K6	4200*	6900

* Estimated by interpolation between temperature of K5 and K7 stars.

M stars are coolest red stars, molecular bands dominate (e.g. TiO), neutral metals, with temperature between 2,000-3,500K. K stars are orange-red stars, with neutral metals (Ca, Fe) dominate, molecular bands appear with temperature between 3,500-5,000K. Star 9 can be either a M3 or M4 star, as its λmax is much higher at around 8200 - 8500 Å. However, uncertainties exist, as its λmax is not available in the data sets. Uncertainties may also arise from the noise, towards the ends of the spectra.

See the figures below for their comparisons.

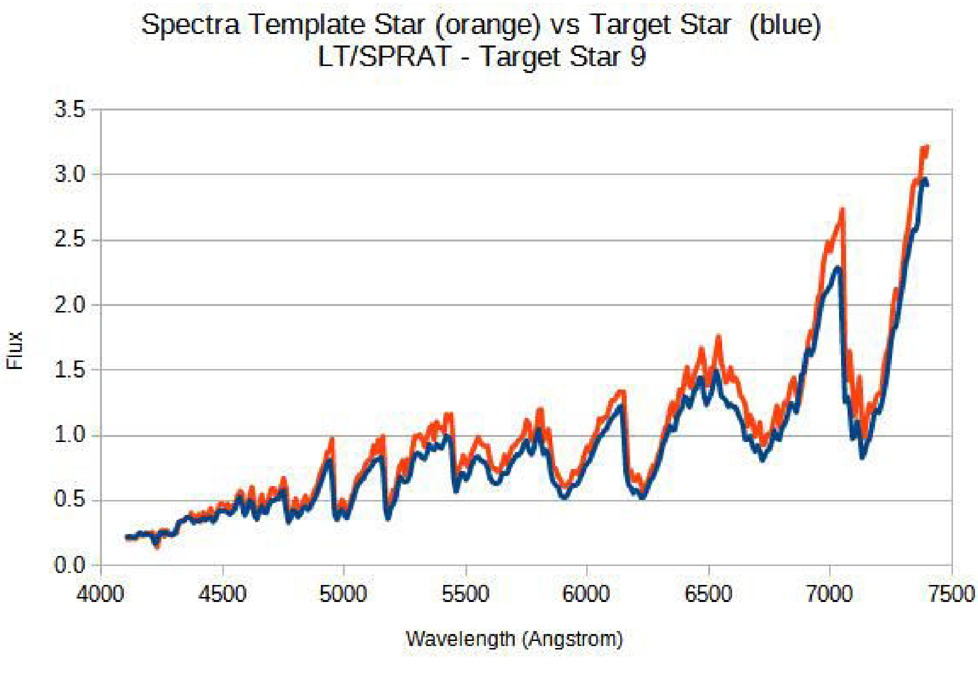

Figure 46: Star 9 vs HD 110964

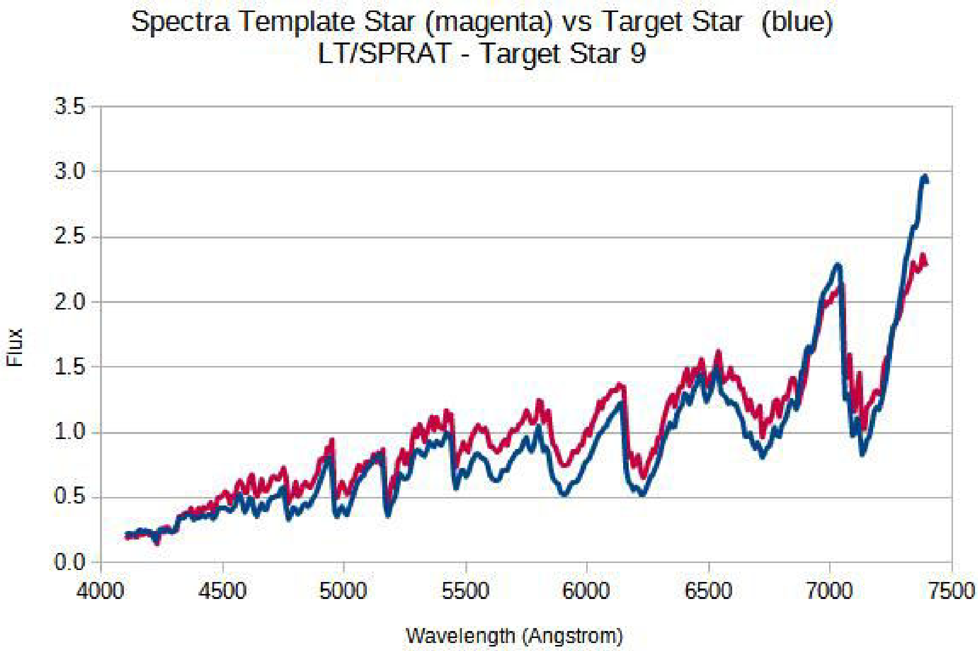

Figure 47: Star 9 vs SAO 63349

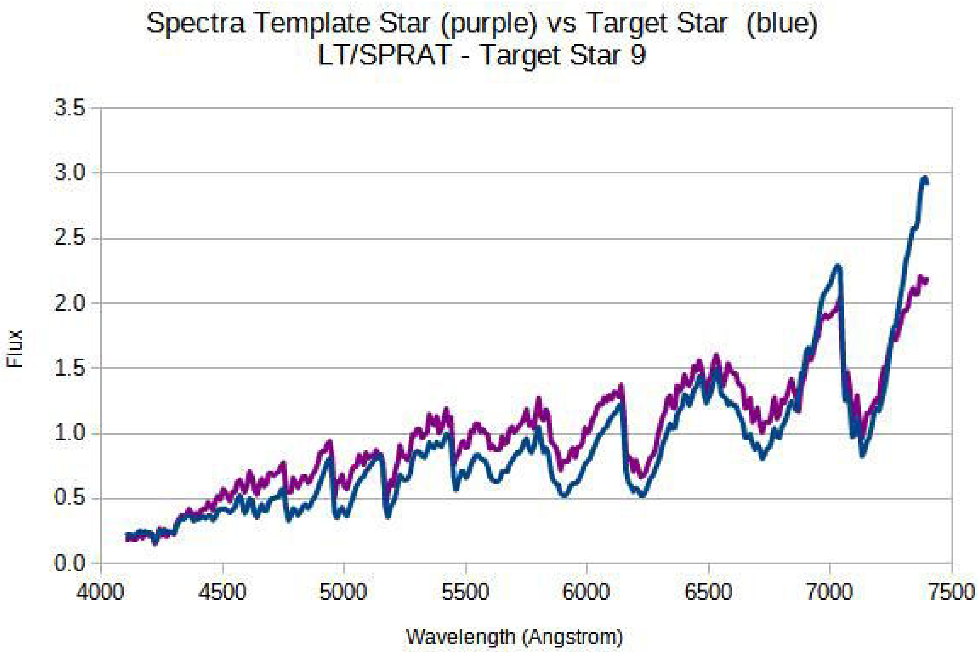

Figure 48: Star 9 vs HD 249826

4 Conclusions:

The template spectra are of real stars and may contain additional absorption or emission lines. There may just be some noise and calibration uncertainties as well. In the classification process, a systematic approach has been used, i.e. "the Least Squares method, instead of just personal judgements.

The wavelength of the data sets only range from 4100Å to 7400Å. So for those elements with shorter wavelength, e.g Ca and other metals, their absorption lines are not available in the classification process. For further improvement, observations with wider ranges of wavelength is recommended.

As a conclusion, the processes of calibration and classification used in this assignment were mainly manual, so was quite time consuming. With the overwhelming astronomical data available these days, the processing and analyzing of data have become challenging. For future improvement, further automation by using software programming, e.g. Python and AI machine learning system are recommended.